e-Hospital AI-Twin: Step-by-Step Interactive App Walkthrough Script

A Click-by-Click, Tab-by-Tab Speaking Guide for Live App Screen Recording

Speaker: Suganthan Subramanian | Faculty of Engineering, University of Ottawa
Format: Screen Recording of the Flutter App (Target Time: 4:15 – 4:30 min | Max: 5 min)

How to use this walkthrough: Open your Flutter app on your phone, simulator, or Chrome browser. Start recording your screen (using Loom, Zoom, or Windows Snipping Tool). Follow the exact **[ON-SCREEN ACTION]** prompts as you speak the words in quotes. Every single tab, button, and module in your app is covered in the exact sequence it appears on screen.

STEP 1: 0:00 – 0:35 | Introduction & Patient Dashboard

[ON-SCREEN ACTION]: Start with the Patient Dashboard visible on screen. Hover over or point to the 'Welcome back, Sugan' banner and the 'Apple HealthKit & Central Cloud Synced' badge.

"Hello judges and evaluators. My name is Suganthan Subramanian, graduate eHealth and biomedical engineering researcher at the University of Ottawa. Today, I am giving you a live, interactive walkthrough of our e-Hospital AI-Twin mobile application.

Right here on our main Patient Dashboard, you can see our real-time synchronization status. The system is actively connected to my Apple Watch via Apple HealthKit and directly synchronized with our hospital's Central AWS App Runner cloud database. Every vital sign captured on the wearable feeds into the clinic EHR with zero manual entry."

STEP 2: 0:35 – 0:55 | The WhatsApp Care Card on Dashboard

[ON-SCREEN ACTION]: Point to the green 'Chat with Clinic on WhatsApp' card right below the welcome banner.

"Notice this prominent green card: 'Chat with Clinic on WhatsApp'. For elderly patients or those who find mobile navigation difficult, this provides an instant one-tap bridge to their clinical care team. Patients can simply text or send a voice note on WhatsApp, and our FastAPI webhook automatically parses their blood pressure or heart rate and writes it straight into the central hospital records."

STEP 3: 0:55 – 1:35 | Vitals Screen: Clinical Reference & Apple Watch Sync

[ON-SCREEN ACTION]: Tap on the 'Vitals & Apple Watch' grid card to navigate to the Vitals Screen. Point to the top 'Clinical Reference' card, then point to the 'Sync Apple Watch' button in the top app bar.

"Now, let's tap into 'Vitals & Apple Watch'.

At the top, the app displays our Clinical Reference card pulled directly from the hospital's central database: showing my latest hospital ECG status, baseline heart rate of 72 bpm, and baseline blood pressure.

Up in the app bar, we have our live 'Sync Apple Watch' button. When tapped, it automatically harvests background biometrics from HealthKit. Next to it is the manual logging button for patients who check their blood pressure at home using cuff monitors."

STEP 4: 1:35 – 2:30 | Vitals Screen: Exploring the 5 Biometric Tabs

[ON-SCREEN ACTION]: Click sequentially through the 5 tabs: Steps -> Calories -> Heart Rate -> Sleep -> Stress.

"Now let's explore the five biometric tabs:

• Tab 1: Steps: Displays interactive daily step counts with selectable range filters for 7, 14, or 30 days, accompanied by automated AI summaries.

• Tab 2: Calories: Tracks active caloric expenditure and metabolic burn curves.

• Tab 3: Heart Rate: Shows continuous cardiac telemetry plotted over time against my hospital baseline. If abnormal spikes occur, the AI automatically alerts the clinician.

• Tab 4: Sleep: Displays nocturnal sleep hours paired with our integrated Sleep Analysis View, tracking sleep architecture and restorative cycles.

• Tab 5: Stress: This is our autonomic recovery tracker. It uses resting heart rate variability to compute a daily stress score. Patients can even tap any point on the chart to attach a personal note explaining what caused their stress."

STEP 5: 2:30 – 3:20 | Health Assistant: Module Picker & Live AI Chat

[ON-SCREEN ACTION]: Tap the back arrow to return to Dashboard. Tap the purple 'AI Assistant' card. You will see the 'Choose an AI tool' module picker with 4 modules. Tap the first module: 'Chat'.

"Let's head back to the Dashboard and open our AI Assistant.

Here you see our modular AI architecture. Let's launch Module 1: 'Health Chat'.

Notice the top green banner providing another one-tap WhatsApp escalation link to Dr. Patel. Down below, we have pre-configured clinical prompt chips. Let's tap: 'What are my recorded vitals?'

[ON-SCREEN ACTION: Tap the chip 'What are my recorded vitals?' and watch the response appear]

Watch our backend work in real time: our LangGraph engine queries our AWS App Runner MySQL database, retrieves my latest wearable vitals, formats them against clinical guidelines, and returns this personalized, empathetic summary."

STEP 6: 3:20 – 3:55 | Report Analyzer, Nutrition Monitor & WhatsApp Module

[ON-SCREEN ACTION]: Tap the Back arrow in the app bar to return to 'Choose an AI tool'. Point to 'Report Analyze', 'Nutrition Monitor', and then tap 'WhatsApp Care Team'.

"Going back to our AI modules, we also have:

• Report Analyze: Which allows patients to upload blood work or clinical PDFs for automated biomarker extraction and plain-language translation.

• Nutrition Monitor: Where patients track meals, receiving real-time dietary scoring customized for hypertension and diabetes.

• Module 4: WhatsApp Care Team: When tapped, it explains our automated Whisper voice-transcription pipeline and provides an 'Open Clinic WhatsApp' button that directly launches a pre-filled chat with Dr. Patel."

STEP 7: 3:55 – 4:25 | Conclusion & System Impact

[ON-SCREEN ACTION]: Return to the main Dashboard or switch your camera full screen.

"In conclusion, what you have seen today is a fully working, closed-loop healthcare system:
1. It captures continuous telemetry from Apple Watches.
2. It reaches every patient through zero-barrier WhatsApp messaging.
3. It saves clinicians up to 2 hours of documentation daily through ambient voice AI and automated triage.

Every component is live, running on AWS App Runner, and built to restore human connection to modern healthcare.

Thank you so much!"